

CCT College Dublin

Assessment Cover Page

Module Title:	Algorithms & Constructs Software Development Fundamentals
Assessment Title:	System Modelling & Build
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Declaration

By submitting this assessment, I confirm that I have read the CCT policy on Academic Misconduct and understand the implications of submitting work that is not my own or does not appropriately reference material taken from a third party or other source. I declare it to be my own work and that all material from third parties has been appropriately referenced. I further confirm that this work has not previously been submitted for assessment by myself or someone else in CCT College Dublin or any other higher education institution.

Merge Sort

The assignment asked for a recursive sorting algorithm, so I used merge sort to sort the employee names in alphabetical order.

Dividing the list into small parts and then joining the parts again in the correct order, I chose this method because I think it works better with larger lists.

Another point for I had chosen merge sort is because it also uses recursion in a clear way, which helped me understand recursive algorithms better.

Binary Search

I used binary search to search employees by name after the list is sorted.

Checking the middle of the list first, if the employee is not there, it ignores half of the list and continues searching only in the correct side. This makes the search faster than checking every employee.

Binary Tree

I used a binary tree to create the employee hierarchy and it connects the nodes using left and right children as required in the assignment.

ArrayList

I used arrayList to store the employees while the program is running.

I used arrayList because it is simple and flexible and allows adding, sorting and searching employee records easily. Another good point is the size can grow automatically when new employees are added.

Conclusion

I used, as examples for my project, some java already available on moodle.

I did a simple project using what I learnt and my brother helped me to understand merge sort, binary tree, root, left and right. Also, I had some help from AI chat gpt.

I used chat gpt to help me to understand some parts of the project. I used it in portuguese.

GPT

1. me explica enum - netbeans

https://chatgpt.com/s/t_6a10acf56a808191afe4ccc0e934ae4a

2. eu crio primeiro essa classe com o enum ou ela vem depois te tudo feito?

https://chatgpt.com/s/t_6a10ad8d0c508191ae6615098c691190

3. por que ta dando errado?

https://chatgpt.com/s/t_6a10afc93f8481918a0f56e0c53375de

4. me explique arraylist int i, j, k

https://chatgpt.com/s/t_6a10b248f4bc8191ad6ccb281a81157e

5. quero por um switch para acessar enum do departamento (add, search, ...). e outro switch para executar cada case/enum. funciona ne?

https://chatgpt.com/s/t_6a10b77748908191ae00867fc4dca51f